F18CD Mark Scheme - December 2023

1. (a) An ordered field $\mathbb{F}$ is said to be complete if for every non-empty subset $A \subset \mathbb{F}$ bounded above, there exists a supremum of A in $\mathbb{F}$.

(b)

Proof. Letting $z = y/x$, it suffices to prove that for every $z \in \mathbb{R}^+$, there exists an $n \in \mathbb{N}$ such that $z < n$. Assume towards a contradiction that $\mathbb{N}$ is bounded above . Then by the completeness axiom $\sup(\mathbb{N})$ exists. Let $\sup(\mathbb{N}) = \alpha$. Then there exists an $m \in \mathbb{N}$ such that

$$\begin{aligned} &\alpha - 1 < m \\ &\Rightarrow \alpha < m + 1. \end{aligned}$$

But, $m + 1 \in \mathbb{N}$ and hence α is not an upper bounded for $\mathbb{N}$. Contradiction. □

(c)

Proof. Clearly A is non-empty as $\mathbb{N}$ is non-empty. Since $5 > 5 - 2/n$ for all $n \in \mathbb{N}$, it follows that 5 is an upper bounded for A. Define (a_n) to be the sequence given by

$$a_n = 5 - \frac{2}{n}.$$

Then $a_n \in A$ for each $n \in \mathbb{N}$ and $\lim(a_n) = 5$. Hence, $\sup(A) = 5$. Since $5 \notin A$, it follows that 5 is not a maximum. □

(d) This statement is true.

Proof. Since $B \subset \mathbb{R}$ is bounded above, $\sup(B)$ exists. Let $\sup(B) = \alpha$. Suppose towards a contradiction that $\alpha \notin B$. Then there exists a $b \in B$ such that

$$\alpha - 1 < b < \alpha.$$

Since $\alpha \notin B$, there exists a $c \in B$ such that

$$b < c.$$

Now, it cannot be that

$$b < c < b + 1$$

and hence $\alpha < b + 1 \leq c$. Contradiction. □

2.(a) A sequence (a_n) converges to $a \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon$$

for every $n > N$.

(b)

Proof. Given $\varepsilon > 0$, there exists an $N_1 \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon/2$$

for every $n > N_1$ and there exists an $N_2 \in \mathbb{N}$ such that

$$|b_n - b| < \varepsilon/2$$

for every $n > N_2$. Let $N := \max(\{N_1, N_2\})$. Then for every $n > N$, it follows that

$$\begin{aligned} |(a_n + b_n) - (a + b)| &\leq |a_n - a| + |b_n - b| \\ &< \varepsilon/2 + \varepsilon/2 \\ &= \varepsilon. \end{aligned}$$

□

(c) Here, $\lim(a_n) = 1$ and $\lim(b_n) = 1/2$.

Proof. $((a_n) \to 1)$ Given $\varepsilon > 0$, by the Archimedean property there exists an $N \in \mathbb{N}$ such that $2/N < \varepsilon$. Then, for every $n > N$, it follows that

$$\begin{aligned} \left| \frac{n}{n+2} - 1 \right| &= \frac{2}{n+2} \\ &< \frac{2}{n} \\ &< \varepsilon. \end{aligned}$$

□

Proof. $((b_n) \to 1/2)$ Given $\varepsilon > 0$, by the Archimedean property there exists an $N \in \mathbb{N}$ such that $6/N < \varepsilon$. Then, for every $n > N$, it follows that

$$\begin{aligned} \left| \frac{n^2 - 3n}{2n^2 - 1} - \frac{1}{2} \right| &= \frac{|-6n + 1|}{|2(2n^2 - 1)|} \\ &< \frac{6n}{2n^2 - 1} \\ &< \frac{6n}{n^2} \\ &< \varepsilon. \end{aligned}$$

□

(d) No, a counterexample is $a_n = 1/n$ and $b_n = n$. Then $(a_n b_n) = (1, 1, 1, \dots)$, which converges to 1.

3.(a) A series $\sum_{k=1}^{\infty} a_k$ converges to L if the sequence (s_n) of partial sums

$$s_n = \sum_{k=1}^{n} a_k$$

converges to L.

(b)

Proof. Suppose $\sum_{k=1}^{\infty} |a_k|$ converges . Since

$$0 \leq a_k + |a_k| \leq 2|a_k|$$

for each $k \in \mathbb{N}$, it follows by the comparison test, and algebra of convergent series, that

$$\sum_{k=1}^{\infty} a_k + |a_k|$$

is convergent. Now, since $\sum_{k=1}^{\infty} |a_k|$ and $\sum_{k=1}^{\infty} a_k + |a_k|$ are convergent, it follows that

$$\sum_{k=1}^{\infty} (a_k + |a_k|) - |a_k| = \sum_{k=1}^{\infty} a_k$$

is convergent.

□

(c)

Proof. Here, $a_k = \cos(k)/k^4$. Since $|a_k| \leq 1/k^4$, it follows by the comparison test and p-series test that $\sum_{k=1}^{\infty} |a_k|$ is convergent. Hence, by part (a), $\sum_{k=1}^{\infty} a_k$ is convergent. □

(d)

Proof. Let $a_k = (-1)^k(\sqrt{k+1} - \sqrt{k})$. Then the sequence of partial sums (s_n) for the series $\sum_{k=1}^{\infty} |a_k|$ is given by

$$\begin{aligned} s_n &= \sum_{k=1}^{n} \sqrt{k+1} - \sqrt{k} \\ &= \sqrt{n+1} - 1. \end{aligned}$$

Since $\lim(\sqrt{n+1}) = \infty$, it follows that (s_n) diverges and hence $\sum_{k=1}^{\infty} |a_k|$ diverges. Now,

$$0 < \frac{1}{\sqrt{k+1} + \sqrt{k}}$$

for each $k \in \mathbb{N}$. Defining $b_k = \sqrt{k+1} + \sqrt{k}$, it is clear that $b_{k+1} \geq b_k$ for all $k \in \mathbb{N}$, since $\sqrt{k+2} \geq \sqrt{k}$. Hence, $|a_{k+1}| \leq |a_k|$ for all $k \in \mathbb{N}$. That is, $(|a_k|)$ is monotone decreasing. Finally, since

$$\lim \left(\frac{1}{\sqrt{k+1} + \sqrt{k}} \right) = 0$$

it follows by the alternating series test that $\sum_{k=1}^{\infty} a_k$ is convergent. □

4.(a) A function $f : D \to \mathbb{R}$ is continuous at $c \in D$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(c)| < \varepsilon$$

whenever $|x - c| < \delta$ and $x \in D$.

(b)

Proof. Let f be differentiable at $c \in D$. Then,

$$\lim_{x \to c} \left(\frac{f(x) - f(c)}{x - c} \right) = L \in \mathbb{R}.$$

Hence,

$$\begin{aligned} \lim_{x \to c} (f(x) - f(c)) &= \lim_{x \to c} \left(\frac{f(x) - f(c)}{x - c} \right) \lim_{x \to c} (x - c) \\ &= L \cdot 0 \\ &= 0 \end{aligned}$$

Hence, $\lim_{x \to c} f(x) = f(c)$ and thus $f(x)$ is continuous at the point $c \in D$. □

(c)

Proof. Given $\varepsilon > 0$, let $\delta = \varepsilon/3$. Then if $0 < |x - 2| < \delta$ it, follows that

$$\begin{aligned} |f(x) - 12| &= \left| \frac{3x^2 - 12}{x - 2} - 12 \right| \\ &= |3(x + 2) - 12| \\ &= 3|x - 2| \\ &< 3\delta \\ &< \varepsilon. \end{aligned}$$

Hence $\lim_{x \to 2}(f(x)) = 12 = f(2)$. □

(d) This statement is false. One counter-example is the function $f : [-2/\pi, 2/\pi] \to \mathbb{R}$ with

$$f(x) = \begin{cases} \sin(1/x) : x \neq 0 \\ 0 : x = 0 \end{cases}$$

This function is not continuous at $x = 0$ but for every $\alpha \in (-1, 1)$ there exists a $c \in (-2/\pi, 2/\pi)$ such that $f(c) = \alpha$.

5. (a) For a power series $\sum_{k=0}^{\infty} a_k x^k$, the radius of convergence$R$ is either

$$R := \sup\left(\left\{|x_0| \in \mathbb{R} : \sum_{k=0}^{\infty} a_k x_0^k \text{ converges } \right\}\right),$$

or ∞ if the series converges for every $x_0 \in \mathbb{R}$.

(b) Let $x_0 \in \mathbb{R}$ and define $a_k = x_0^k/k$. Then,

$$\begin{aligned} \lim_{k\to\infty} \left| \frac{x_0^{k+1}}{k+1} \frac{k}{x_0^k} \right| &= \lim_{k\to\infty} \left| \frac{kx_0}{k+1} \right| \\ &= |x_0| \lim_{k\to\infty} \left| \frac{k}{k+1} \right| \\ &= |x_0|. \end{aligned}$$

By the ratio test, $\sum_{k=1}^{\infty} x_0^k/k$ converges if $|x_0| < 1$ and diverges if $|x_0| > 1$. Hence the radius of convergence is 1.

(c) Each $a_k = f^{(k)}(0)/k!$ by Taylor's theorem. Hence,

$$a_k = \frac{(-1)^{k+1}}{k}$$

for $k \geq 1$ and $a_0 = 0$. When $x = 1$, this gives the alternating harmonic series, which is convergent . When $x = -1$, this gives the harmonic series but with each term multiplied by -1. Since the harmonic series is divergent, this is also divergent.

(d) This is false. Let $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} e^{-x^{-2}} & : x \neq 0 \\ 0 & : x = 0 \end{cases}$$

Then $f^{(k)}(0) = 0$ for each $k \in \mathbb{N}$. If

$$f(x) = \sum_{k=1}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$$

for all $x \in \mathbb{R}$ then $f(x) = 0$. This is clearly not $f(x)$.